Chapter 4 Multivariate Normal Population Statistical Inference

4.1. There is a certain relationship between the amount of sweating and the content of potassium（钾）in human body. The amount of perspiration (出汗(x_1)) of 20 healthy adult women was measured. The content of sodium (x_2) and potassium content (x_3) were also measured. The data are listed in the following table. It is assumed that the data $\mathrm{x} = (x_1, x_2, x_3)'$ obeys a trivariate normal distribution.

试验者	x_1	x_2	x_3
1	3. 7	48. 5	9. 3
2	5. 7	65. 1	8. 0
3	3. 8	47. 2	10. 9
4	3. 2	53. 2	12. 0
5	3. 1	55. 5	9. 7
6	4. 6	36. 1	7. 9
7	2. 4	24. 8	14. 0
8	7. 2	33. 1	7. 6
9	6. 7	47. 4	8. 5
10	5. 4	54. 1	11. 3
11	3. 9	36. 9	12. 7
12	4. 5	58. 8	12. 3
13	3. 5	27. 8	9. 8
14	4. 5	40. 2	8. 4
15	1. 5	13. 5	10. 1
16	8. 5	56. 4	7. 1
17	4. 5	71. 6	8. 2
18	6. 5	52. 8	10. 9
19	4. 1	44. 1	11. 2
20	5. 5	40. 9	9. 4

It is known that the sample mean vector $\bar{x} = (4.64, 45.4, 9.965)'$ and

$$\boldsymbol{S} = \begin{bmatrix} 2.88 & 10.01 & -1.81 \\ 10.01 & 199.79 & -5.64 \\ -1.81 & -5.64 & 3.63 \end{bmatrix}$$

$$\boldsymbol{S}^{-1} = \begin{bmatrix} 0.586 & -0.022 & 0.258 \\ -0.02 & 0.006 & -0.002 \\ 0.258 & -0.002 & 0.401 \end{bmatrix}$$

.

Please answer the following questions.

(1) Construct a Hotelling T^2 statistic to test the hypothesis

$$H_0: \mu = \mu_0 = (4,50,10)', H_0: \mu \neq \mu_0 \ (\alpha = 0.05),$$

where $F_{0.05}(3,17) = 3.2$.

(2) Find the 95% confidence region of μ.

4.3 Using the data in the table 4.4.1, test the following hypotheses of the female baby population at the significance level $\alpha = 0.05$.

(1) $H_0: \mu = (80, 60, 15)', H_1: \mu \neq (80, 60, 15)'$;

(2) $H_0: \frac{1}{5}\mu_1 = \frac{1}{4}\mu_2 = \mu_3$,

H_1: there are at least two terms in $\frac{1}{5}\mu_1, \frac{1}{4}\mu_2, \mu_3$ that are unequal.

表 4.4.1　　　　某地区农村女婴的体格测量数据

编号	身高(cm)	胸围(cm)	上半臂围(cm)
1	80	58.4	14.0
2	75	59.2	15.0
3	78	60.3	15.0
4	75	57.4	13.0
5	79	59.5	14.0
6	78	58.1	14.5
7	75	58.0	12.5
8	64	55.5	11.0
9	80	59.2	12.5

4.4 There are two brands of tires, A(甲) and B(乙). Now we take 6 tires for durability test. The test is carried out in three stages. The first stage rotates 1000 times, the second stage rotate 1000 times, and the third stage rotate 1000 times as well. The durability indicators are listed in the table below.

Phase A(甲)			Phase B(乙)		
1	2	3	1	2	3
194	192	141	239	127	90
208	188	165	189	105	85
233	217	171	224	123	79
241	222	201	243	123	110
265	252	207	243	117	100
269	283	191	226	125	75

Under the assumption of multivariate normality, are there significant differences in the durability indicators between A and B ($\alpha = 0.05$)? If so, which stage is different?

4.10 A prison divides prisoners into three parts: ordinary prisoners, crazy prisoners and other prisoners. 20 prisoners were selected from each of the three parts to measure the length of their ears. Under the hypothesis of multivariate normality, we tried to test whether there was significant difference in the length of the ears of the three parts（ α= 0.05）。 The data are listed in the table below.

普通犯人			疯狂犯人			其他犯人		
测量对象	左耳	右耳	测量对象	左耳	右耳	测量对象	左耳	右耳
1	59	59	1	70	69	1	63	63
2	60	65	2	69	68	2	56	57
3	58	62	3	65	65	3	62	62
4	59	59	4	62	60	4	59	58
5	50	48	5	59	56	5	62	58
6	59	65	6	55	58	6	50	57
7	62	62	7	60	58	7	63	63
8	63	62	8	58	64	8	61	62
9	68	72	9	65	67	9	55	59
10	63	66	10	67	62	10	63	63
11	66	63	11	60	57	11	65	70
12	56	56	12	53	55	12	64	64
13	62	64	13	66	65	13	65	65
14	66	68	14	60	53	14	67	67
15	65	66	15	59	58	15	55	55
16	61	60	16	58	54	16	56	56
17	60	64	17	60	56	17	65	67
18	60	57	18	54	59	18	62	65
19	58	60	19	62	66	19	55	61
20	58	59	20	59	61	20	58	58

Additional exercise. Suppose $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n$ is a sample of population $\boldsymbol{x} \sim N_p$ $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}>0$. Consider the following hypothesis problem:

$$H_0: \mu = \mu_0 \quad vs \quad H_1: \mu \neq \mu_0$$

Problem:

1) Derive the likelihood function of sample $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n$, which can be formulated as

$$L(\mu, \Sigma)=(2\pi)^{-np/2}|\Sigma|^{-n/2}\exp\{tr[-\frac{1}{2}\Sigma^{-1}(A + n(\bar{x}-\mu)(\bar{x}-\mu)']\}.$$

2) Let $\Theta = \{(\mu, \Sigma): \mu \in R^p, \Sigma > 0\}$ and $\Theta_0 = \{(\mu_0, \Sigma): \mu_0 \in R^p, \Sigma > 0\}$. Then, the maximum likelihood estimators of μ, Σ are $\bar{x}$ and $\mathrm{A/n}$, respectively. Please prove that the likelihood ratio statistic (LRT) can be formulated as

$$\lambda = \frac{max_{\Theta_0} L(\mu,\Sigma)}{max_{\Theta} L(\mu,\Sigma)} = (1 + T^2/(n-1\})^{-n/2},$$

where $T^2 = n(\bar{x}-\mu_0)'S^{-1}(\bar{x}-\mu_0)$ is the Hotelling T^2 statistic.

3) Note that λ is a strictly increasing function of T^2 and the rejection rule of LRT λ is $\lambda \leq \lambda_0$ for critical value $0 < \lambda_0 < 1$ of likelihood ration test. Therefore, the rejection rule of T^2 should be T^2_____c such that $P(T^2 \geq c) = \alpha$, where α is the level of Hotelling test.

Please submit your assignments on next Tuesday (10: 00, April 2, 2026).